

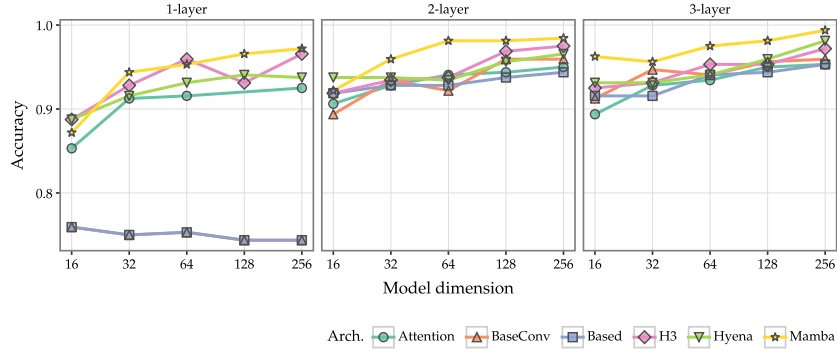


Figure 13: **Sibling queries:** Accuracy across models with 1–3 layers on ATR ($L_{\max} = 5$, $|\Sigma| = 20$)

models, even more so than parent queries, and thus does not provide useful signal for comparing architectures.

Why are sibling queries easy? Parent nodes are guaranteed to be special terminals in our GNF which are sampled from a nonterminal-dependent distribution (see §B). However, siblings have a large chance of being fixed terminals specified by the production rule. Additionally, the rightmost sibling of a particular terminal may be itself, if it is the rightmost terminal of its production rule. We speculate that these factors combined make sibling queries easier than parent queries, and thus not a suitable testbed for multihop reasoning.

Future work. The appropriate analogue to study multihop *reasoning* in ATR is grandparent relations (or higher up ancestors in the tree), since the grandparent is always a special head terminal (like the parent) and is always to the right of the parent and thus different from the query terminal. We leave further experiments on this to future work.